Minimum Viable Product (MVP) Definition and Rationale

Ecosystem Discovery Application

1. Introduction

The Minimum Viable Product (MVP) represents the smallest functional version of the Ecosystem Discovery Application that delivers meaningful educational value to users.

The purpose of defining an MVP is to identify the core system features required to demonstrate the main learning interaction of the application while keeping the development scope manageable for the project timeline.

For this project, the MVP focuses on a hint-based interactive discovery game within a freshwater ecosystem, where students identify animals using clues, collect them in a sticker book, and reinforce learning through visual interaction and feedback.

2. MVP Vision

The envisioned MVP for the Ecosystem Discovery Application provides a single interactive freshwater ecosystem environment combined with a structured game-based learning experience.

When users enter the game scene, they see a Saskatchewan-inspired freshwater habitat with animated animals. Instead of freely clicking animals, users interact through a hint-based card system. Each mystery card represents a hidden animal.

The user selects a card, reads progressively revealed hints, and then identifies the correct animal within the ecosystem. Correct identification unlocks the animal in the sticker book, while incorrect attempts reveal additional hints or result in failure.

The system also includes a countdown timer to introduce challenge and engagement, along with optional bonus interactions (such as collecting trash for extra time).

This structured interaction encourages active thinking, curiosity, and engagement while supporting Grade 3 ecosystem learning outcomes.

3. Core MVP Features

The MVP includes the following essential features:

Feature	Description
Interactive Game Scene	A visual Saskatchewan freshwater ecosystem with animated animals
Hint-Based Discovery System	Users select mystery cards and identify animals using progressively revealed hints

Feature	Description
Timed Gameplay	A countdown timer introduces challenge and engagement during gameplay
Sticker Book System	Correctly identified animals unlock collectible stickers
Hint System	Multiple hints are revealed step-by-step for each animal
Visual Feedback System	Correct and incorrect actions trigger animations and visual feedback
Bonus Interaction	Users can gain extra time through interactions such as trash collection
Educational Video Support	A videos page provides additional learning through embedded educational content
Dynamic Animal Data	Animal and hint data are fetched from a backend database (Supabase)
MVC Architecture	System structured using Model–View–Controller design with modular JavaScript

4. MVP Interaction Flow Diagram

User Opens Website
↓
Home / Start Screen
↓
Enter Game Page
↓
User Presses Start
↓
Timer Begins (120 seconds)
↓
User Selects Mystery Card
↓
Hint Displayed
↓
User Clicks Animal in Scene
↓
Correct → Sticker Unlocked → Progress Updated
↓
Wrong → Next Hint OR Card Fails
↓
Repeat Until All Cards Complete OR Timer Ends

↓
End Screen Displayed (Results)

5. MVP User Experience Flow

The MVP interaction flow is designed to be simple, engaging, and intuitive for young learners.

1. The user opens the Ecosystem Discovery Application.
2. The home screen introduces the learning experience.
3. The user navigates to the game page.
4. The user presses the Start button to begin the game.
5. A countdown timer begins.
6. The user selects a mystery card representing a hidden animal.
7. The system displays a hint related to the selected animal.
8. The user clicks an animal in the ecosystem scene.
9. If correct:
   - The card flips and the sticker is unlocked
   - Progress is updated
10. If incorrect:
    - The system provides feedback and reveals additional hints
    - If all hints are used, the card is marked as failed
11. The user continues until all cards are completed or time expires.
12. The system displays an end screen with results.

6. MVP Rationale

The selected MVP features represent the minimum functionality required to deliver the educational objective of the system while incorporating interactive gameplay elements.

These features allow students to:

- Engage with a structured and interactive learning experience
- Identify animals using hints and critical thinking
- Learn ecological information through guided discovery
- Receive visual rewards through the sticker collection system
- Reinforce learning through repetition and feedback

By focusing on a hint-based game, timer system, and visual feedback, the MVP enhances engagement compared to simple click-based exploration while remaining manageable within the project scope.

This approach ensures that the system is both educational and engaging, while demonstrating key software engineering concepts such as MVC architecture, state management, and backend data integration.